
Layerwise Signals for Transformer MLP Compression

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Abstract

We study whether layerwise statistics in pretrained Transformer MLPs can guide post-training low-rank compression. Across the four chosen small language models, we measure centered effective rank, MLP output energy, and Gaussian mutual information between each layer and final layer on WikiText-2 activations. The descriptive analysis shows clear nonuniform structure: normalized effective-rank and output-energy signals are strongly anticorrelated across layers, and Gaussian mutual information shows that individual MLP outputs have a flat early/middle regime followed by a sharp late-layer rise for the contrution of information to the final output. Together with effective rank, the MI results suggest a late-layer compression-informativeness tradeoff: late MLP outputs become more aligned with the final representation while using fewer effective dimensions. We then test a simple activation-aware layerwise rank budget built from rank and energy signals. Activation-aware factorization is much stronger than plain SVD at the same rank budget, but the tested layerwise allocations do not beat a uniform activation-aware allocation. The main conclusion is that Transformer MLPs contain interpretable layer structure, but converting that structure into a better compression policy requires more direct sensitivity modeling than the simple additive budget used here.

1 Introduction

Transformer MLP blocks contain a large share of language-model parameters, making them natural targets for pre-training low-rank compression. Uniform rank allocation is simple, but it assumes that all layers should receive the same compression budget. This project asks whether layerwise statistics reveal useful structure for deciding where rank should be preserved.

We study four small causal language models and focus on three layer signals: effective rank, MLP-output energy, and Gaussian mutual information between the intermediate and the final layer. The project has two goals. First, we describe how these signals vary across depth. Second, we test whether a simple rank/energy budget improves activation-aware low-rank compression over a uniform budget. The work connects low-rank adaptation and compression ideas [1, 2] with effective-rank and Information Bottleneck style representation analysis [3–5]. Our main finding is mixed: layerwise structure is strong and interpretable, but the tested layerwise compression rule does not beat activation-aware uniform allocation.

2 Methods

Models and data. We evaluate four pretrained causal language models: Qwen3-0.6B [6], Qwen2.5-1.5B [7], Llama-3.2-1B [8], and SmolLM2-1.7B [9]. All statistics and benchmarks use WikiText-2 [10]. Layer statistics are collected from 65,536 calibration tokens. Perplexity benchmarks use fixed token windows and the same tokenization and batching protocol for base and compressed variants.

Layer statistics. For each layer ℓ , we collect the MLP input, the MLP output, and the residual stream immediately before the MLP input normalization. The main dimensionality statistic is centered effective rank. For activation matrix A , with covariance eigenvalues λ_i ,

$$\tilde{r}_i = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}.$$

We divide by hidden dimension to obtain an effective-rank ratio. The rank signal is the average of the MLP-input and MLP-output centered effective-rank ratios. The energy signal is the log uncentered trace of the MLP output,

$$\tilde{e}_i = \log \text{tr}\left(\frac{1}{n} A^\top A\right) = \log\left(\frac{1}{n} \sum_{i=1}^n \|A_i\|_2^2\right).$$

We also compute Gaussian mutual information between each layer’s MLP output m_ℓ and the final layer’s MLP output m_L , with residual-stream MI as a cumulative reference. For MLP outputs, the primary estimator uses CCA:

$$I_{\text{CCA}}(m_\ell; m_L) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2),$$

where ρ_i are canonical correlations after truncated whitening. In this report, effective rank is a proxy for representation complexity and Gaussian MI with the final layer is a proxy for output informativeness.

Rank-budget rule. The layerwise compression rule min-max normalizes the rank signal $\tilde{r}_i$ and energy signal $\tilde{e}_i$ within each model, then defines a protection score

$$s_i = w\tilde{r}_i + (1 - w)\tilde{e}_i.$$

For each target factorized/dense parameter ratio $\alpha \in \{0.5, 0.8\}$ and $w \in \{0.25, 0.5, 0.75\}$, the same total rank budget as a uniform allocation is redistributed across layers in proportion to s_i . We compress only `mlp.gate_proj` and `mlp.up_proj`; `mlp.down_proj` is left dense.

Compression methods. We compare three variants to separate the effect of the approximation rule from the effect of the rank budget. The baseline is *plain SVD uniform*: each target matrix is replaced by a truncated SVD with one rank shared across layers. *Activation-aware uniform* keeps the same uniform ranks, but uses the saved MLP-input second moments to reduce the calibration objective $\mathbb{E}_x \|Wx - \hat{W}x\|_2^2$. In practice, this factorizes the weighted matrix $WE[xx^\top]^{1/2}$ and maps the rank- k approximation back with $E[xx^\top]^{-1/2}$. *Activation-aware layerwise* uses the same activation-aware approximation, but replaces the uniform rank with the layerwise ranks from the protection score.

3 Results

Rank and energy disagree. Figure 1 shows that normalized effective rank and MLP-output trace vary systematically across depth. The relationship is negative in every model, as summarized in Table 1. Layers that look high-dimensional by centered effective rank often have lower MLP-output energy, while lower-rank layers can write with higher energy. This suggests that a single notion of “compressibility” is not enough to describe MLP layer roles.

Uniform activation-aware compression remains the benchmark. Plain SVD uniform degrades perplexity severely, while activation-aware uniform compression is much stronger at the same rank budget. However, the simple layerwise rank rule does not improve over activation-aware uniform. Even the best layerwise setting for each model and compression level has positive ΔNLL in Table 1. The Qwen models have some mild losses, but Llama-3.2-1B and SmolLM2-1.7B degrade substantially.

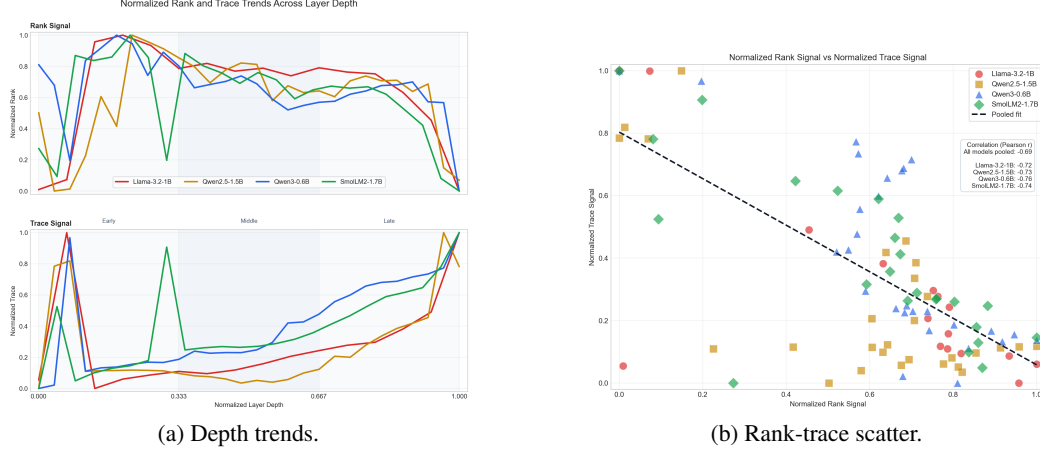


Figure 1: Layerwise EDA. Normalized effective-rank and MLP-output trace signals vary with depth and show a strong negative relationship across models.
Table 1: Compact quantitative summary. Left: Pearson correlation between normalized rank and trace signals. Right: best activation-aware layerwise result relative to activation-aware uniform, reported as Δ average NLL; all values are positive, so uniform remains better.

Rank-trace correlation	
Model	r
Pooled	-0.695
Llama-3.2-1B	-0.723
Qwen2.5-1.5B	-0.730
Qwen3-0.6B	-0.764
SmolLM2-1.7B	-0.740

Best layerwise Δ NLL vs. uniform		
Model	$\alpha = 0.5$	$\alpha = 0.8$
Llama-3.2-1B	2.313	1.776
Qwen2.5-1.5B	0.530	0.056
Qwen3-0.6B	0.074	0.142
SmolLM2-1.7B	2.255	1.995

Mutual information separates local and cumulative layer behavior. Figure 2 compares Gaussian MI for two representations across normalized depth. The MLP-output curve measures the information in each layer’s individual MLP contribution relative to the final MLP output. The residual-stream curve measures information in the accumulated hidden state relative to the final residual stream. Across all four models, residual-stream MI rises smoothly and accelerates in the late layers, consistent with information accumulating through the residual pathway. By contrast, MLP-output MI is much less monotone: it stays relatively flat through early and middle layers, shows sharp drops at a few early low-rank layers, and then rises strongly near the end of the network. This contrast suggests that many early and middle MLPs make local transformations that are not yet strongly aligned with the final representation, while late MLPs become increasingly output-aligned.

To quantify this joint behavior, Figure 3 plots information density—Gaussian MI divided by effective-rank ratio—across normalized depth. All four models exhibit a U-shaped curve: density is high at the earliest layers, drops through the early-to-middle regime, and rises sharply in the final layers. The early peak reflects near-rank-1 MLP outputs that pack their small MI into very few effective dimensions, while the late rise shows that final layers concentrate increasing output-relevant information into a declining number of effective dimensions. The flat middle trough corresponds to layers whose representations are neither particularly low-rank nor strongly aligned with the final output. This U-shape suggests that a compression budget should treat the two ends of the network differently: early bottleneck layers are already maximally compressed by the model itself, while late layers carry dense, output-critical information that is costly to discard.

4 Discussion

Interpreting the results. The original compression question receives a mixed answer. Layerwise statistics clearly reveal nonuniform MLP structure, but the simple rank/energy budget tested here does not improve over activation-aware uniform compression. This negative compression result is informative. It suggests that descriptive statistics such as effective rank and output trace are useful for understanding layer roles, but they are not sufficient by themselves to predict the perplexity sensitivity of low-rank approximation. One interpretation is that pretrained models already contain an implicit resource allocation across layers, making a uniform post-training rank budget harder to beat than expected. Another is that the budget formula protects the wrong layers, or protects them for the wrong

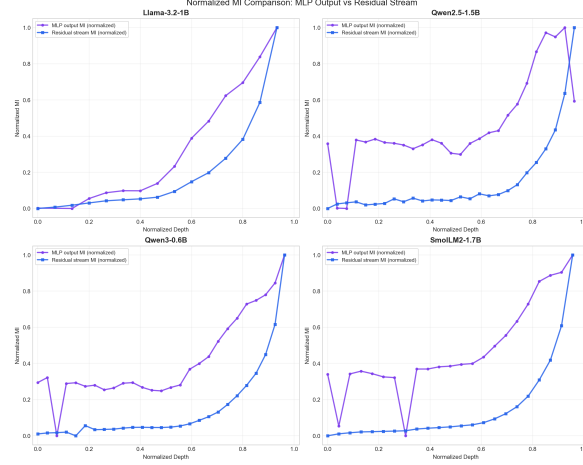


Figure 2: Gaussian mutual information by layer depth. Purple curves show per-layer MLP-output MI computed with CCA; blue curves show residual-stream MI computed with a log-determinant estimator.

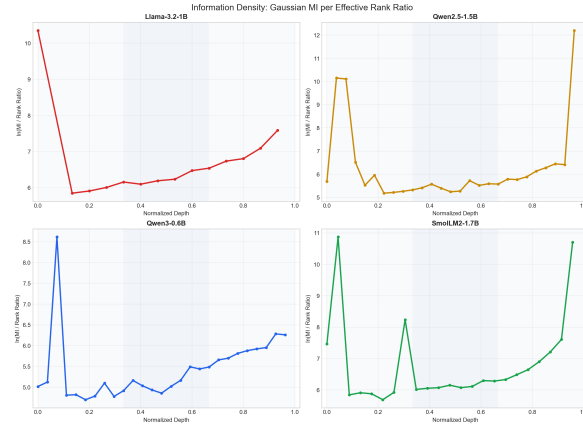


Figure 3: Information density (Gaussian MI per effective-rank ratio) across normalized layer depth. All four models show a U-shaped profile: high density at the earliest and latest layers, with a flat trough in the middle.

reason. The MI results point in this direction: middle-layer MLPs may be weakly aligned with the final representation, early rank-1 layers may act as local bottlenecks rather than simple redundancy, and late layers may be information-dense even when their effective rank is falling. The U-shaped information density profile (Figure 3) reinforces this: a budget that ignores the concentration of output-relevant information at both ends of the network will misallocate rank. This helps explain why a budget based only on rank and trace can miss behavior that matters for compression. A better budget may need to combine representation statistics with direct estimates of layer sensitivity.

Limitations. The study has several limitations. The model suite is limited to 0.6B–1.7B models. All calibration and evaluation uses WikiText-2. Only the MLP gate and up projections are compressed, while the down projection and attention modules are left unchanged. The rank budget is a simple additive heuristic over two normalized signals. Finally, Gaussian MI relies on covariance-based approximations and uses the final layer as a proxy target.

Future study. Future work should estimate per-layer compression sensitivity directly, incorporate MI-derived late-layer protection or information density into the budget, compress the full MLP including down_proj, and test larger models and downstream tasks. The main lesson from this project is not that layerwise statistics are unhelpful. Rather, the statistics expose real structure, and the next step is to design a compression rule that respects that structure more directly.

Data and Code Availability Statement

The project code and generated result artifacts are included in the this repository: [link TBA]. The experiments use WikiText-2 [10] and pretrained Hugging Face model checkpoints for Qwen3-0.6B [6], Qwen2.5-1.5B [7], Llama-3.2-1B [8], and SmolLM2-1.7B [9].

Author Contribution Statement

Peng Zhao developed the rank/trace statistics pipeline, rank-budget computation, model compression scripts, perplexity benchmarking, compression-result analysis, and report integration. Fergie Yang developed the Gaussian mutual-information analysis, information-plane interpretation, MI findings, and MI visualizations. Both authors contributed to the research question brainstorming, presentation design, result interpretation, and report writing.

AI Usage Statement

AI tools were used as implementation and writing assistants. The authors designed the high-level project structure as a deliberately simple staged pipeline. Codex and Claude Code then helped write and revise the individual scripts, debug implementation details, and keep the stage interfaces consistent. Codex and Claude Code also helped with report writeup.

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